

Constraint Satisfaction Problem (CSP)

A constraint satisfaction problem uses a factored representation for each state: a set of variables, each of which has a value. The problem is solved when each variable has a value that satisfies all the constraints on the variable.

Example 1) Which number does each letter represent?

$$\begin{array}{r}
 \text{ T W O} \\
 + \text{ T W O} \\
 \hline
 \text{ F O U R}
 \end{array}
 \quad
 \begin{array}{l}
 \text{T: } \underline{7} \quad \text{W: } \underline{3} \quad \text{O: } \underline{4} \quad \text{F: } \underline{1} \quad \text{U: } \underline{6} \quad \text{R: } \underline{8} \\
 \text{One solution.}
 \end{array}$$

Set the problem

Variables: $\{T, W, O, F, U, R\}$

Domains: $D_i = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ $C_i = \{0, 1\}$

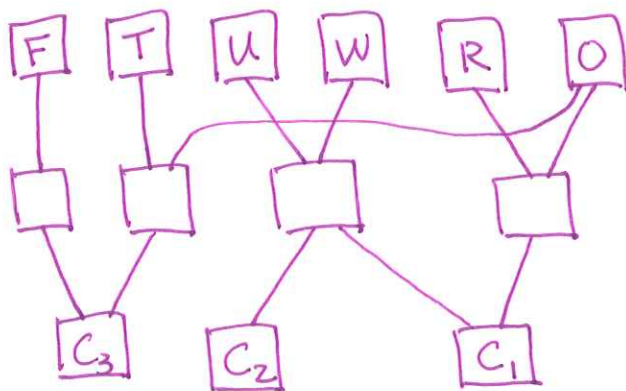
Constraints: All variables have one digit distinct whole numbers.

$$O + O = R + 10 * C_1, \quad C_1 + W + W = U + 10 * C_2,$$

↑
carry over
from 1's digit

$$C_2 + T + T = O + 10 * C_3$$

$$C_3 = F, \quad F \neq 0$$

Constraint Hypergraph

④ Let $O = 4$ and $R = 8$,
 $2T = O + 10 = 14$
 $T = 7$

⑤ $C_1 + W + W = U + 10 * C_2$
 $2W = U$
 $W = \{\cancel{0}, \cancel{1}, \cancel{2}, \cancel{3}, \cancel{4}, \cancel{5}, \cancel{6}, \cancel{7}, \cancel{8}, \cancel{9}\}$
 $W \neq U$ (indicated by arrows pointing to W and U in the set)
 $F \neq 4$ (indicated by an arrow pointing to F in the set)
 U is one digit (indicated by a bracket over the set)
 $\therefore W = 3, U = 6$

Solve:

① $C_3 = \{0, 1\}$, $C_3 = F$, $F \neq 0 \Rightarrow F = 1$

Choose $C_1 = C_2 = 0$ for convenience.

② $C_2 + T + T = O + 10 * 1$

$$2T = O + 10 \Rightarrow T = \{5, 6, 7, 8, 9\}$$

③ $O + O = R + 10 * 0$

$$2O = R$$

choose $O = 4$ to check.

$$O = \{0, 2, 4, 6, 8\}$$

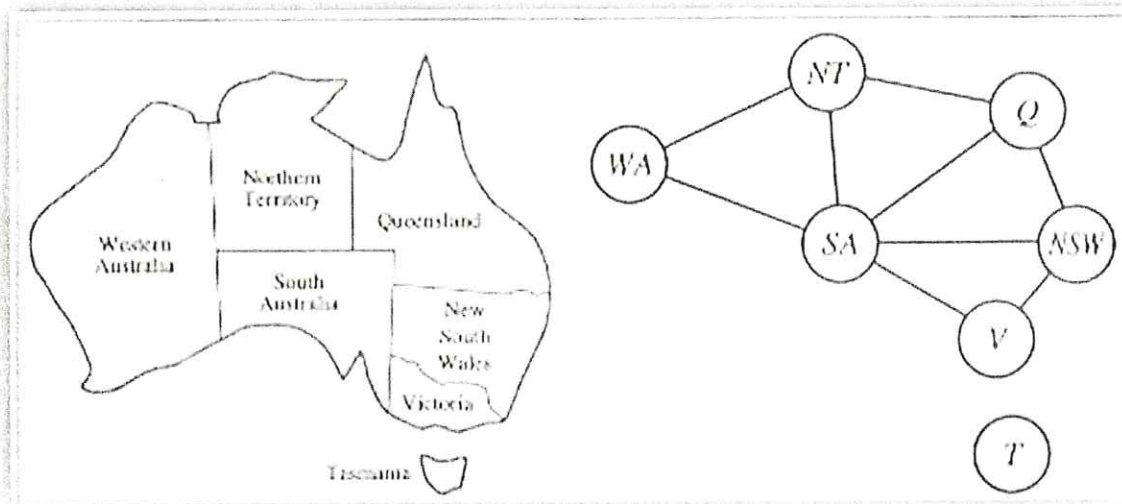
$$\Rightarrow O = \{2, 4\}$$

$$R = \{4, 8\}$$

Techniques:

- **Backtracking Search** is used for a depth-first search that chooses values for one variable at a time and backtracks when a variable has no legal values left to assign.
- **Forward checking:** one of the simplest form of inference. It stops checking whenever a variable does not have any legal value. Constraint propagation and arc consistency.
- **Minimum remaining values (MRV):** Choose the variable with the fewest legal values. When the next unassigned variable is selected, use MRV heuristic.
- **Least constraining value:** Choose the variable that rules out the fewest values in the remaining variables.

Example 2) Map coloring: Below is a map of Australia showing each of its states and territories. You are given the task of coloring each region either red, green, or blue in such a way that no neighboring regions have the same color.



a. Define the variables.

$$\{WA, NT, SA, Q, V, NSW, T\}$$

b. Domain of each variable is:

$$D_i = \{red, green, blue\}$$

c. The constraints require neighboring regions to have distinct colors, so there are nine constraints:

$$\{WA \neq NT, WA \neq SA, NT \neq Q, NT \neq SA, SA \neq Q, SA \neq NSW, SA \neq V, NSW \neq Q, NSW \neq V\}$$

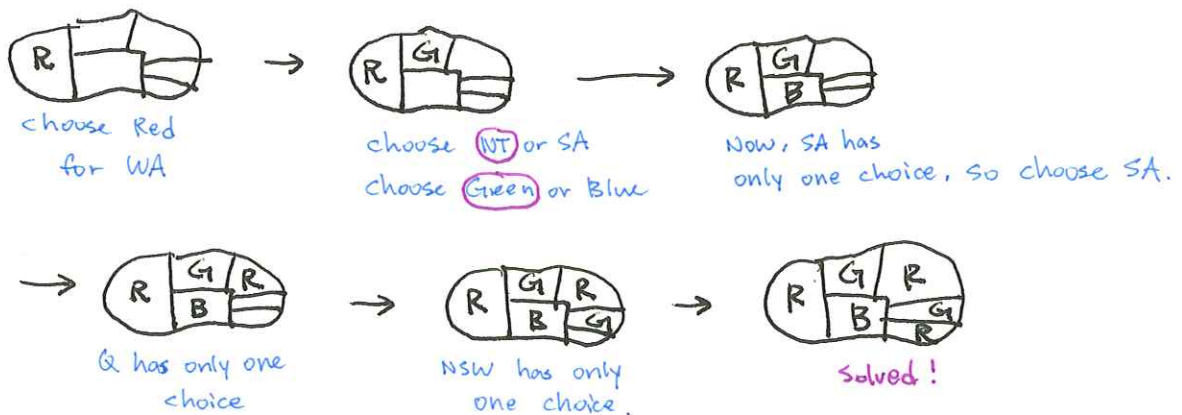
d. Solve: *One example of forward checking and displaying steps.*

Forward Checking:

WA	NT	SA	Q	V	NSW	T
RGB	RGB	RGB	RGB	RGB	RGB	RGB
Choose Red → R	G B	G B	RGB	RGB	RGB	RGB
R	B	B	G ← Randomly choose Green	RGB	R B	RGB
R	B		G	RGB	RB	RGB

↳ becomes empty → stop checking.

Minimum Remaining Values (MRV):



Least Constraining Value:

